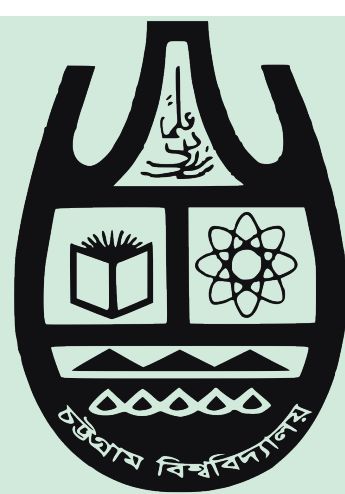




Quantum Computing : Revolutionizing the Future

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Abstract

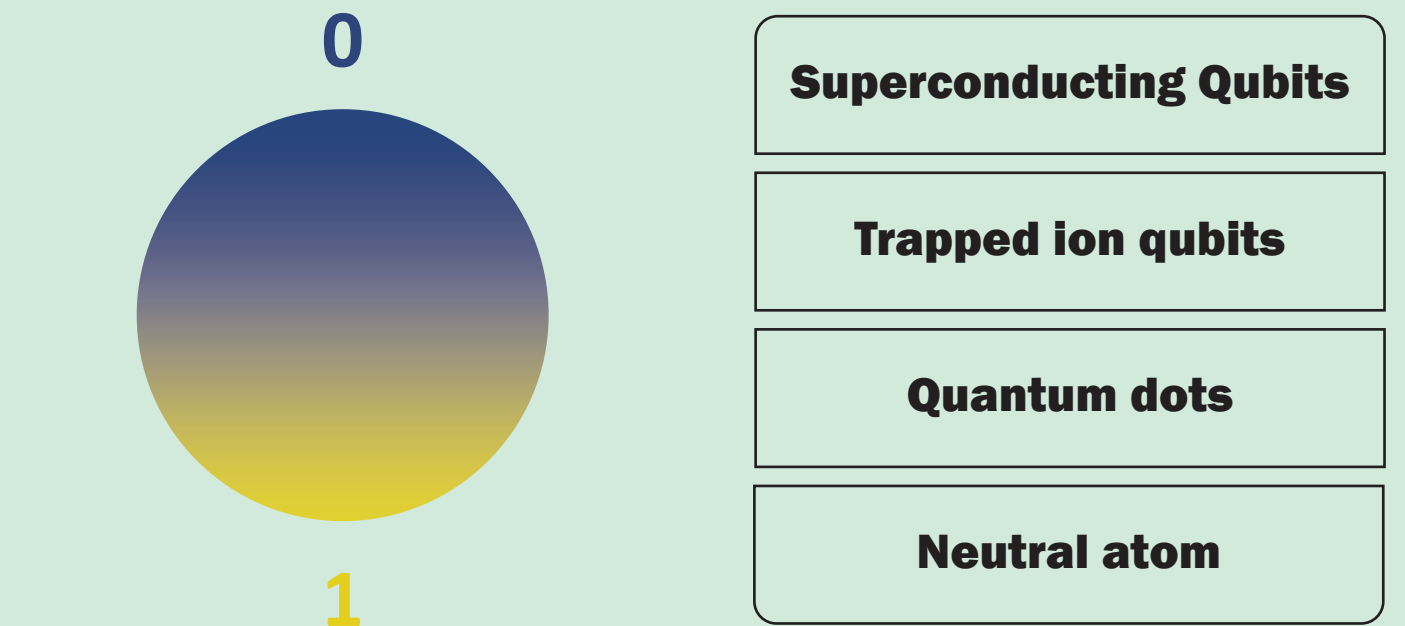
This poster represents innovative visualization techniques designed to simplify the understanding of complex quantum computing by exploring key concepts such as superposition and entanglement that underpin this technology. We highlight current applications from drug discovery to secure data transmission while discussing ongoing challenges such as hardware stability. leveraging interactive diagrams, analogies, and color coding we aim to enhance comprehension among both beginners and experts alike.

Introduction

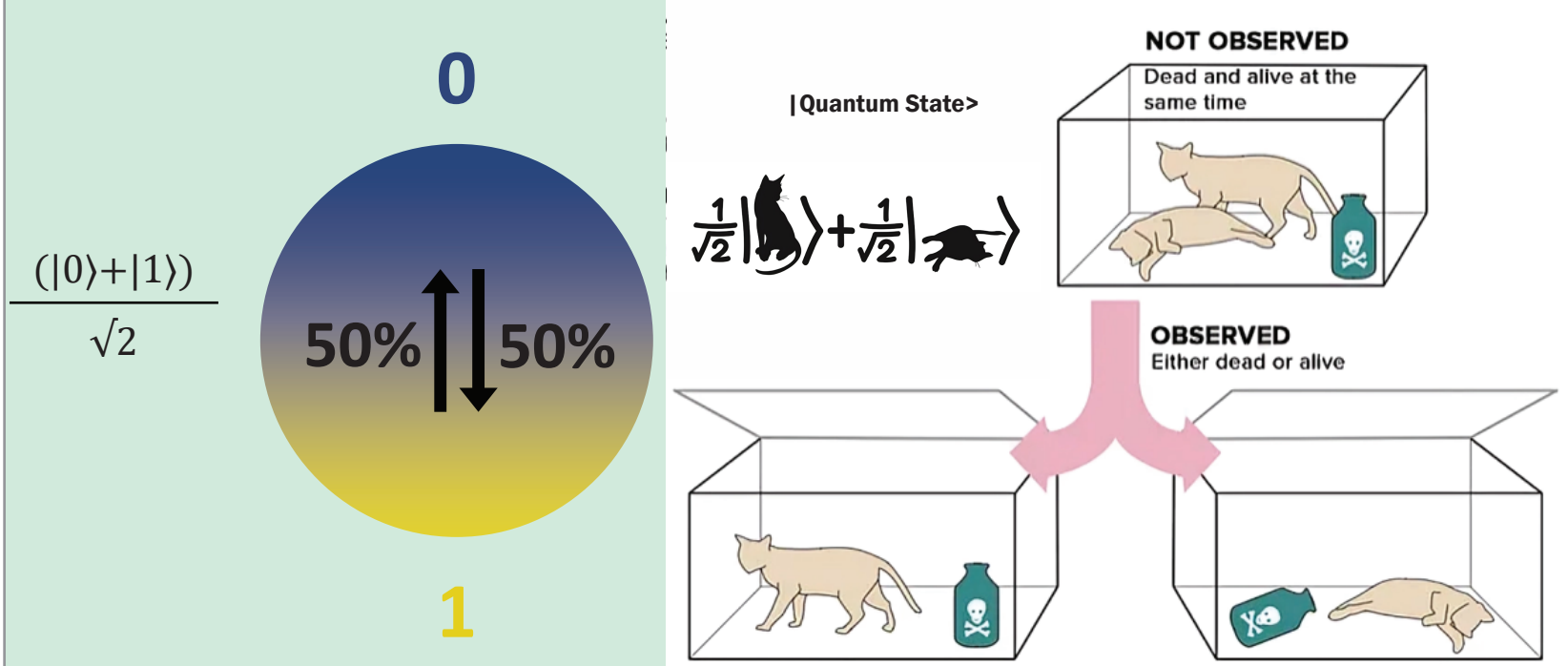
Quantum computation is a multidisciplinary field comprising aspects of computer science, physics and mathematics that utilizes quantum theory to solve complex problems faster than on classical computers. Quantum theory explains the behavior of energy and material on the atomic and subatomic levels. Quantum computing uses subatomic particles, such as electrons or photons.

Key Concepts of Quantum Computing

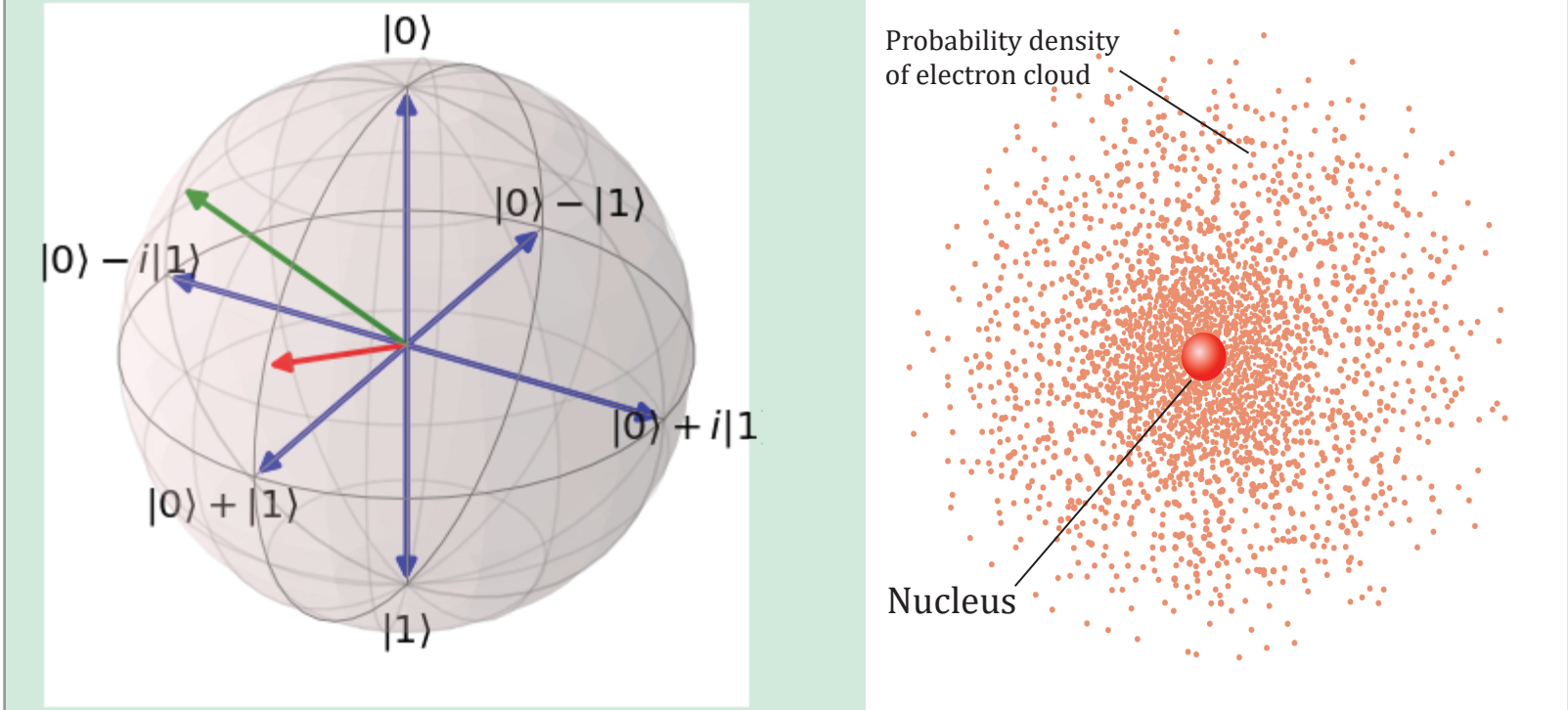
Qubit



Superposition



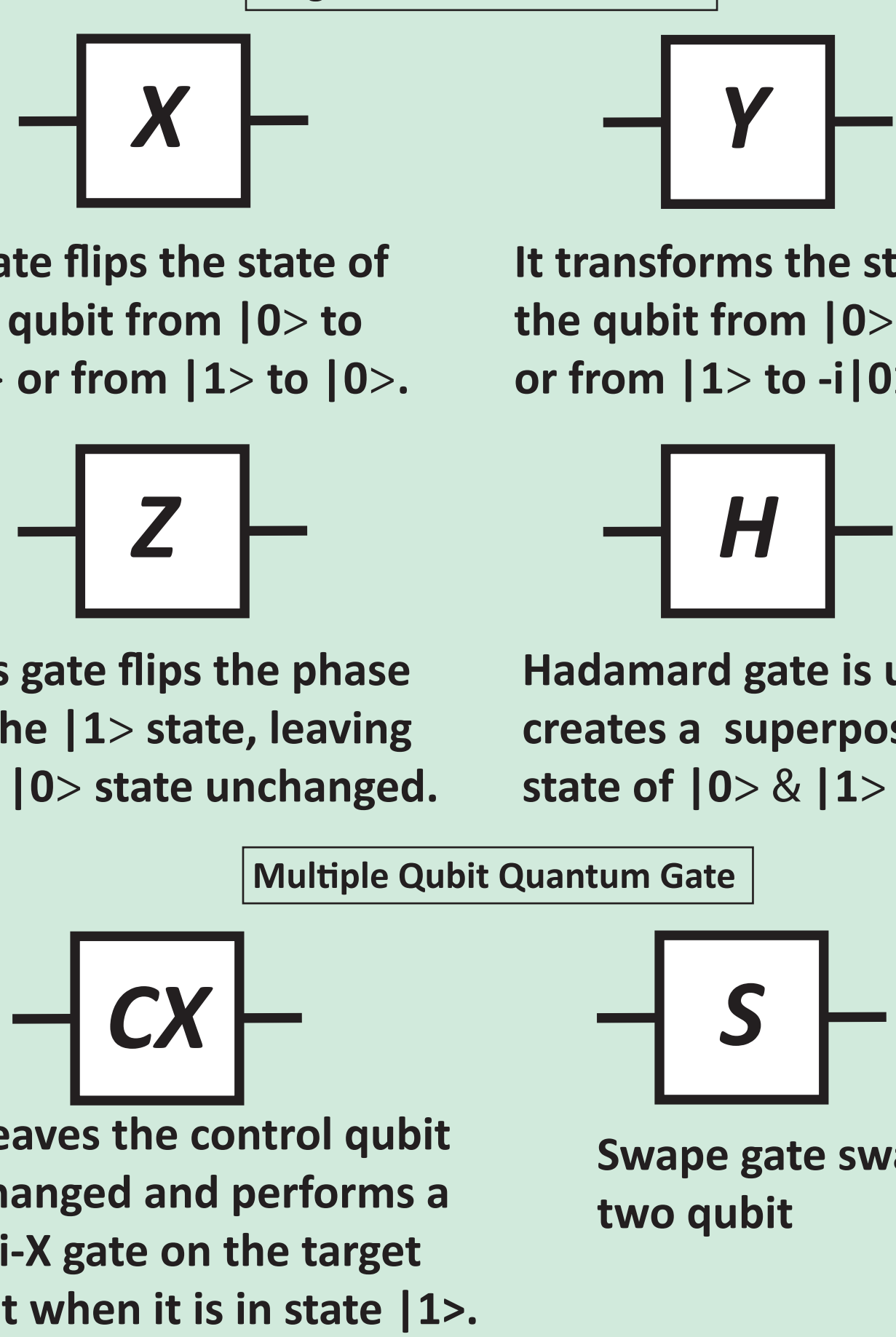
The Bloch sphere provides a useful means of visualizing the state of a single qubit and operations on it. Any point on this sphere represents a linear combination of the 0 and 1 states with complex coefficients.



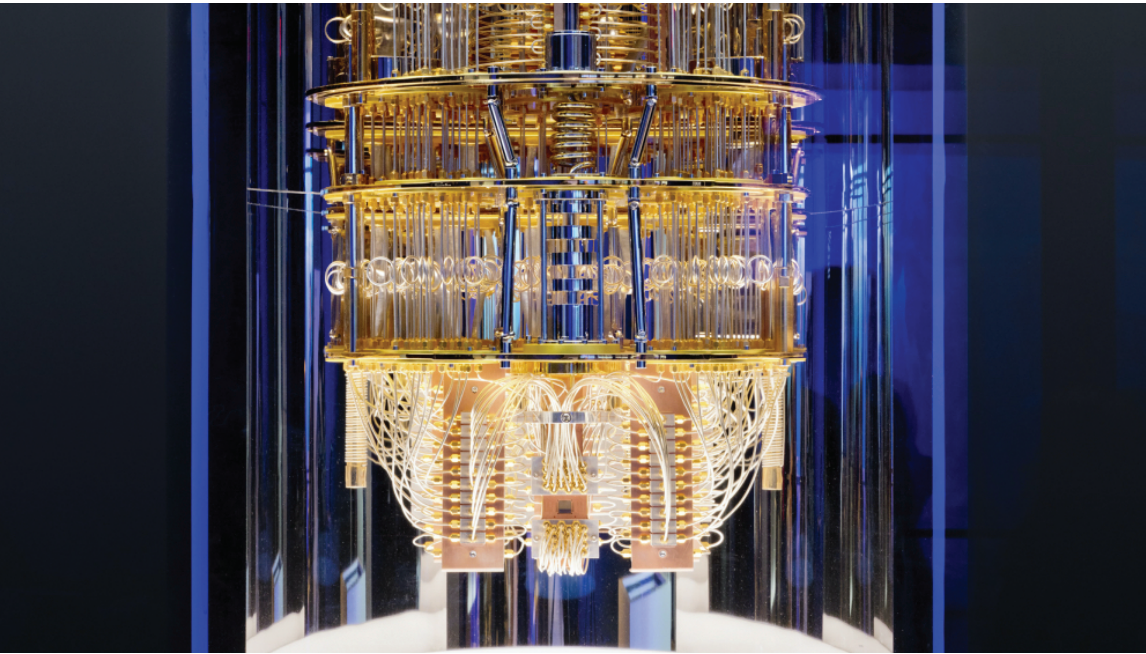
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Quantum Gates

The quantum gate is the building block of quantum computing. Quantum gates represent reversible operations that transform the quantum state according to unitary matrices.



How a Quantum Compter looks like



ibm_sherbrooke

QPU status

● Online

Processor type

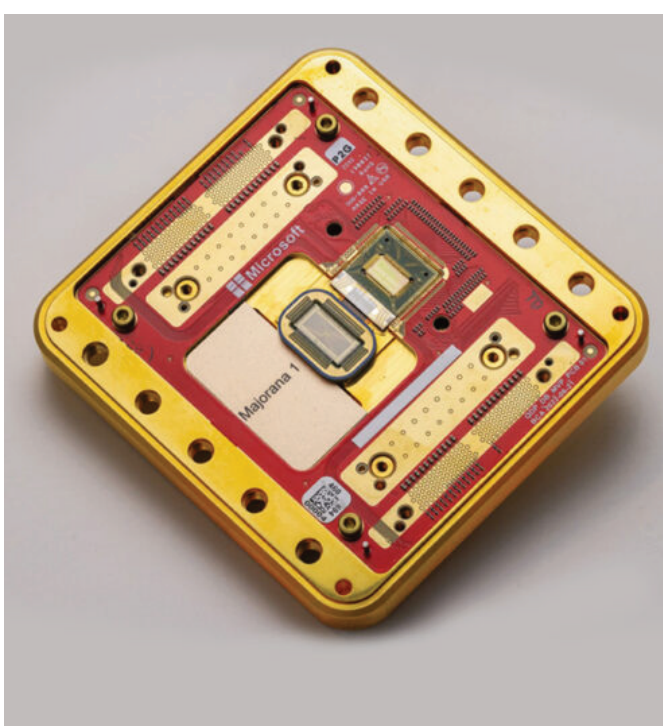
Eagle r3

Qubits	2Q error (best/layered)	CLOPS
127	2.81e-3/2.11e-2	30K

The interior of a quantum computer at the University of Sherbrooke, Canada.

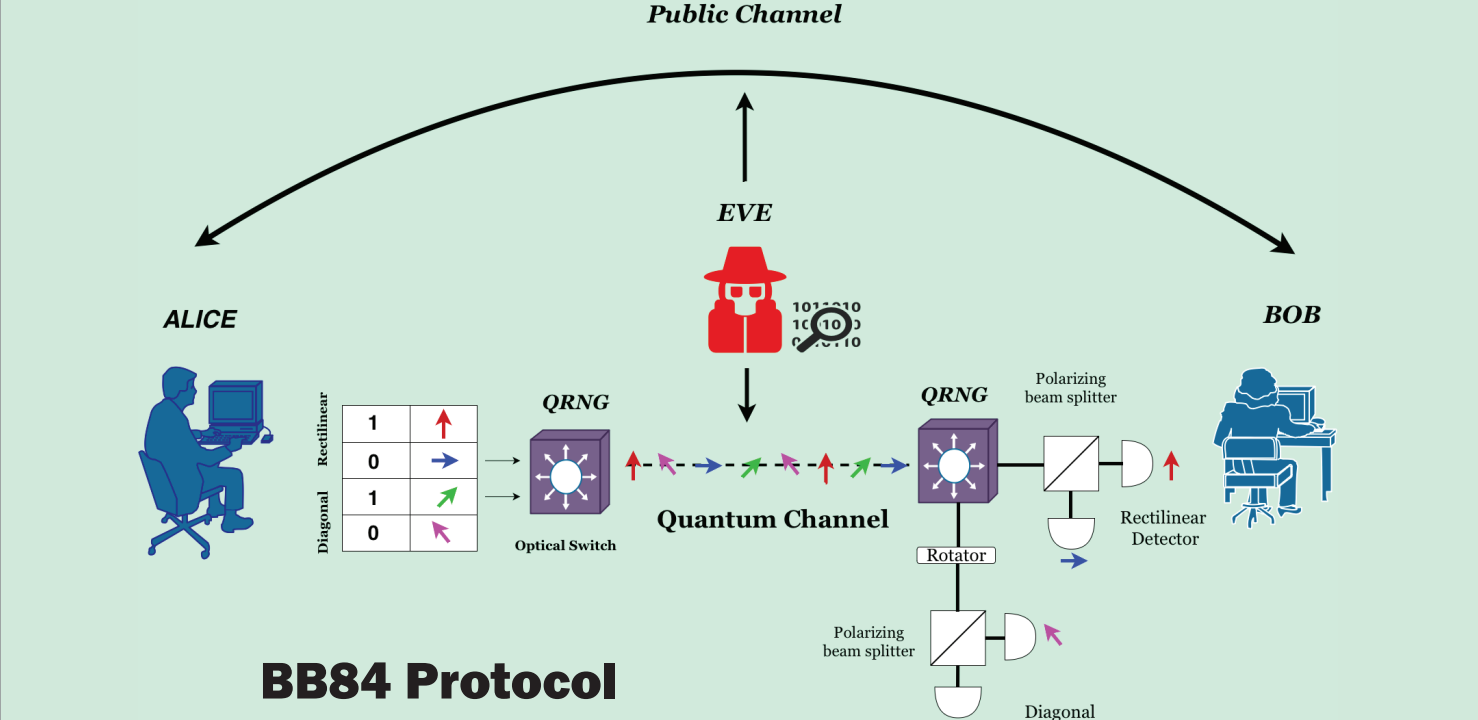


In December 2023, IBM Quantum unveiled Condor, a groundbreaking quantum processor consists of 1,121 superconducting qubits and leveraging IBM Quantum's cross-resonance gate technology.



On 19th Feb 2025, Microsoft reveals its first quantum computing chip. Majorana 1.

Quantum Key Distribution



(QKD) represents a groundbreaking approach to generating secure symmetric encryption keys based on the principles of quantum mechanics. This technique involves the transmission of numerous light particles, known as photons, through fiber optic cables. The sender and receiver assess which photons match to create a secure encryption key. Notably, if an eavesdropper attempts to interfere, the state of the photons changes, alerting both parties to the breach.

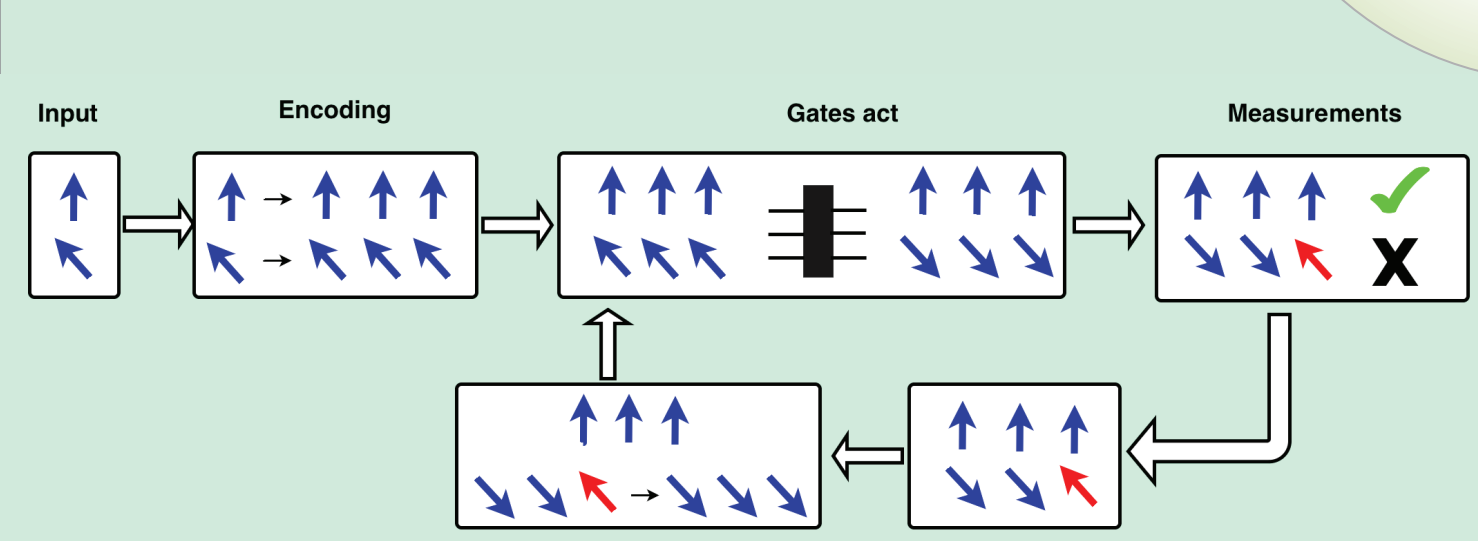
Quantum Algorithm

Quantum algorithms hold the potential to solve certain problems more efficiently than classical algorithms, but they also introduce new challenges, such as managing quantum noise and scaling up to larger systems.

- Shor's Algorithm:** It can factor large numbers exponentially faster which has significant implications for cryptography.
- Variational Quantum Algorithm:** It can be used to calculate molecular energies more accurately which has potential significance in advancing drug discovery.
- Grover's Algorithm:** It can search an unstructured database quadratically, making it useful for optimization problems.

Research Domain

Quantum Error Correction



Quantum error correction (QEC) involves techniques in quantum computing that safeguard quantum information from errors due to decoherence and noise. Information is encoded in multiple physical qubits representing a logical qubit, allowing for error detection and correction. The main error types include bit flips, phase flips, and both. Errors are identified through syndrome measurements on auxiliary qubits that don't disturb the encoded state. Once detected, corrective operations using Pauli matrices (X, Z, or Y) are applied to restore the original state.

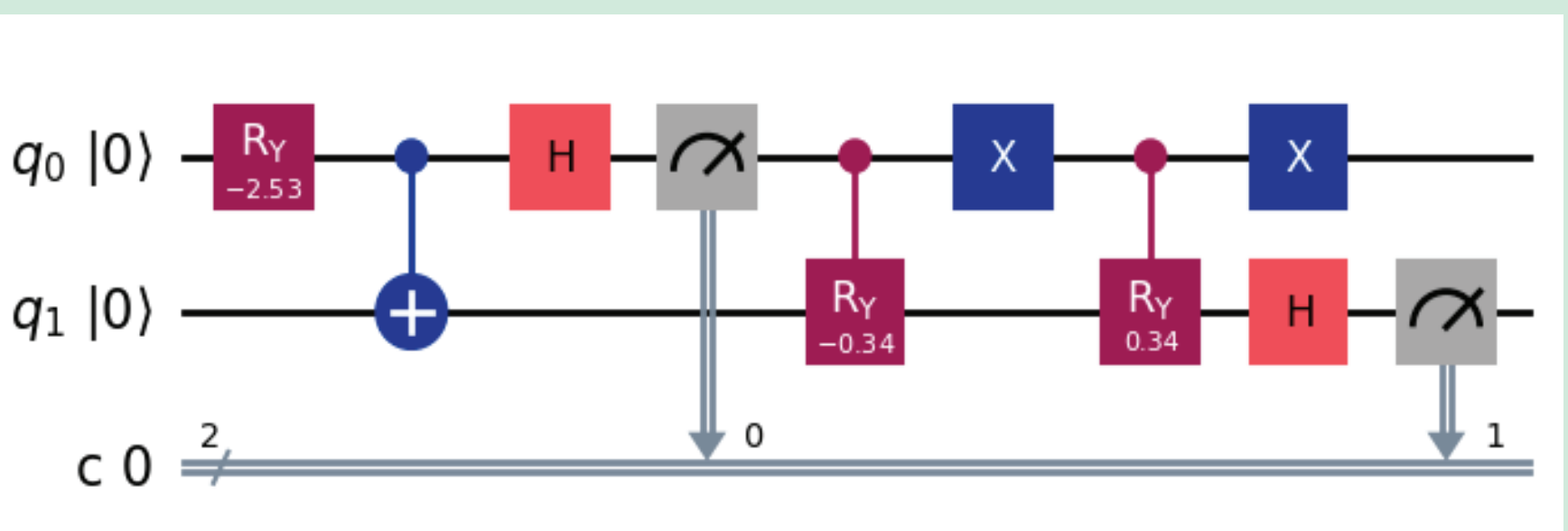
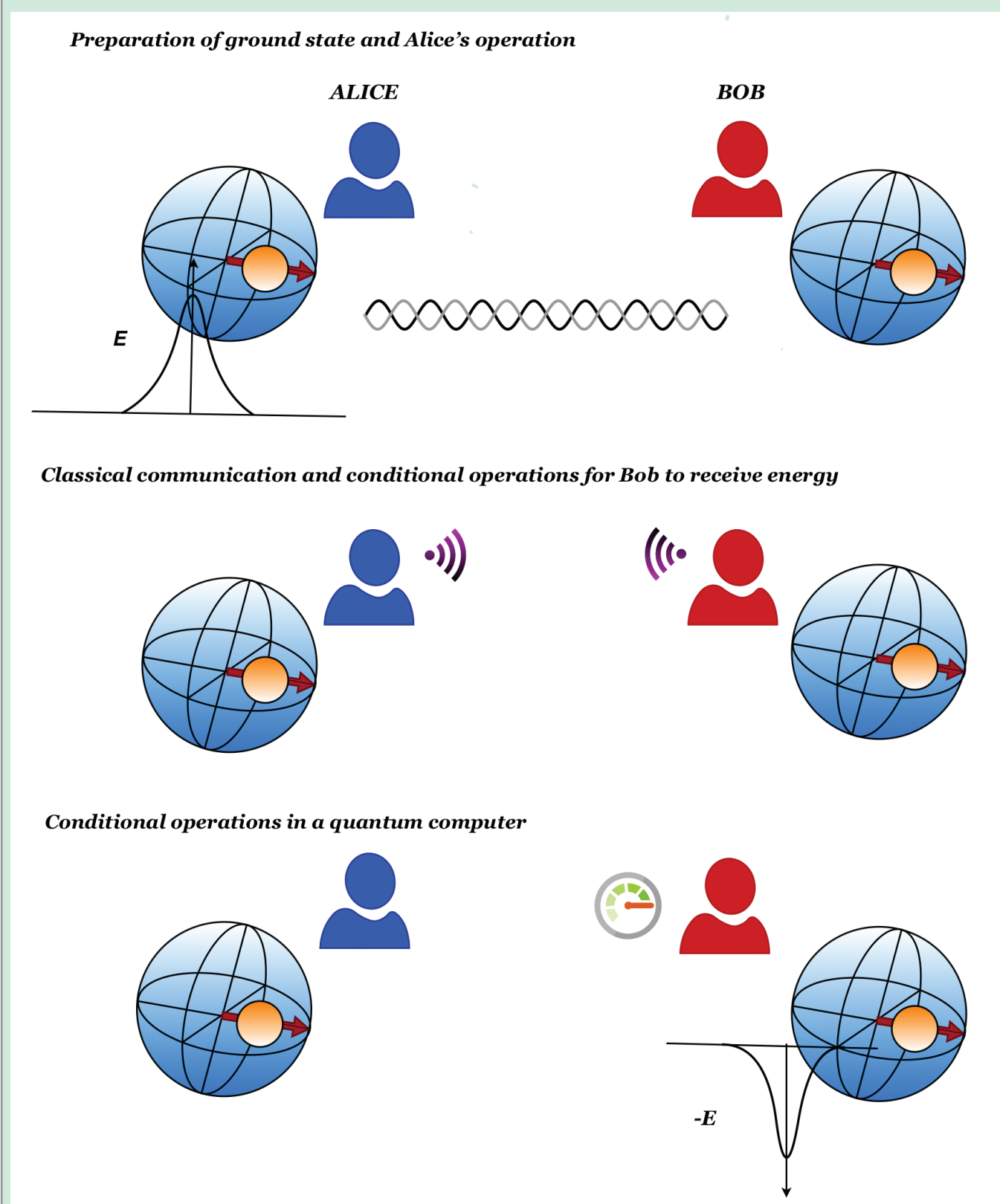
Quantum Machine Learning

Quantum machine learning leverages the information processing power of quantum technologies to enhance and speed up the work performed by a machine learning model. It is also called quantum-enhanced machine learning. Quantum-enabled ones exponentially increase storage and processing power. This ability to store and process huge amounts of information means that quantum computers can analyze massive data sets that would take classical methods significantly longer to perform. Despite these intriguing applications, quantum computing and machine learning are still growing and changing. Some of these applications include to:

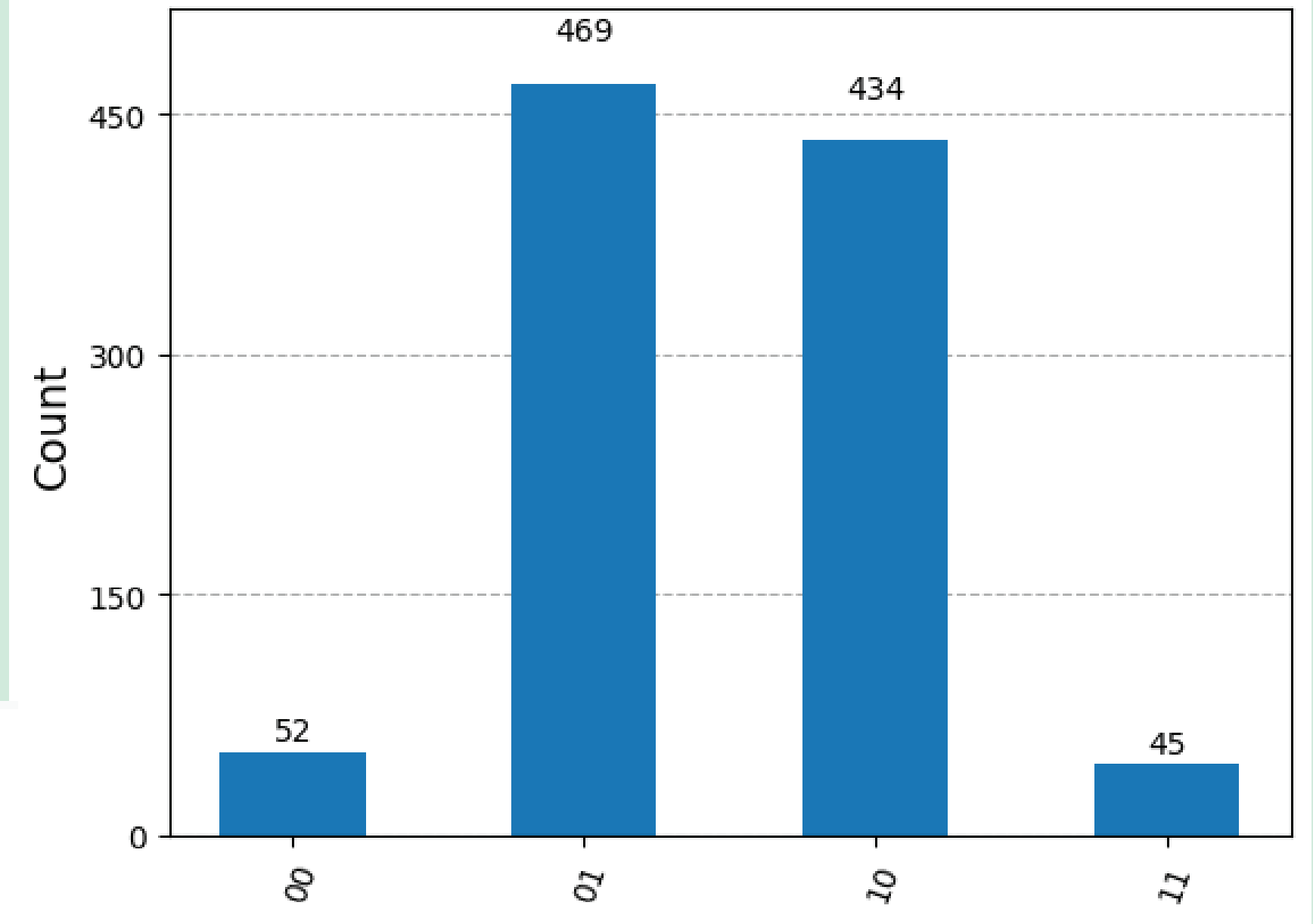
- Develop a new machine learning algorithm.**
- Speed up already existing machine learning algorithms.**
- Create quantum neural networks.**

Quantum Energy Teleportation

Quantum Energy Teleportation (QET) is a novel method that leverages quantum entanglement to transfer energy between two distant locations without any physical movement of the energy. In this phenomenon, locally inaccessible energy is activated as extractable work through collaborative local operations and classical communication (LOCC) with an entangled partner. Although first proposed by Masahiro Hotta in 2008, QET was first realized for a 2-qubit system much later in 2023 by Kazuki Ikeda in a superconducting hardware.



First demonstration of QET protocol established by Ikeda Kazuki in 2023.



Challenges of Quantum Computing

- Qubit Stability
- Error Correction
- Algorithm Development
- Hardware Limitations
- Scalability
- Resource & Cost

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